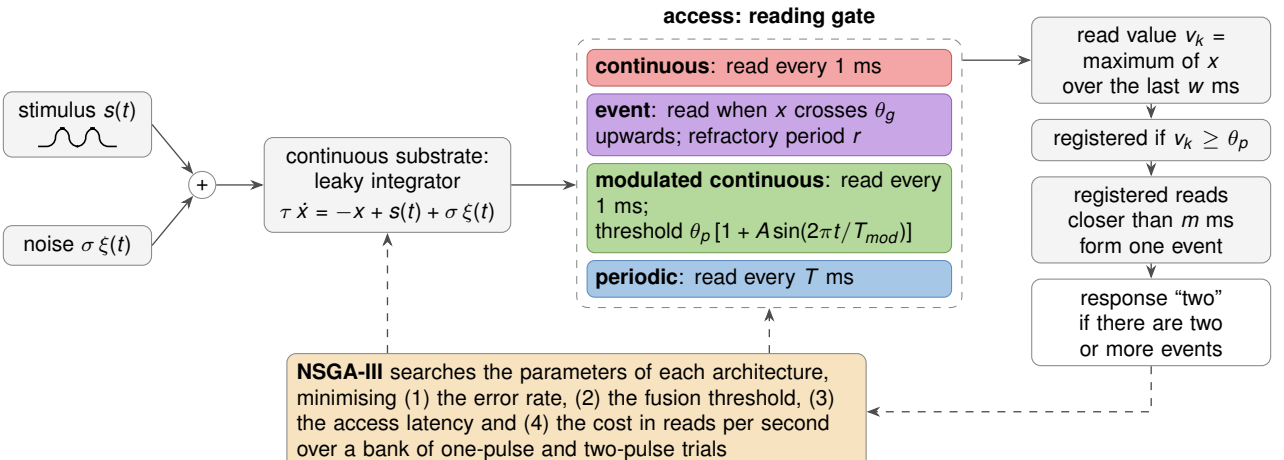
access: reading gate
stimulus $s(t)$
noise $\sigma\,\xi(t)$
+
continuous substrate:
leaky integrator
$\tau\,\dot{x} = -x + s(t) + \sigma\,\xi(t)$
continuous: read every 1 ms
event: read when x crosses θ_g upwards; refractory period r
modulated continuous: read every 1 ms;
threshold $\theta_p\,[1 + A\sin(2\pi t/T_{mod})]$
periodic: read every T ms
read value v_k = maximum of x over the last w ms
registered if $v_k \geq \theta_p$
registered reads closer than m ms form one event
response "two" if there are two or more events
NSGA-III searches the parameters of each architecture, minimising (1) the error rate, (2) the fusion threshold, (3) the access latency and (4) the cost in reads per second over a bank of one-pulse and two-pulse trials